FOUR EQUIVALENT CONDITIONS FOR AN UNRAMIFIED MONOMORPHISM OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2866-2880 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 2882-2903, the complete author proof (20 body lines).

Editorial note. Navigation stubs: 5 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Unramified morphisms of algebraic spaces, monomorphisms, universal injectivity, and formal unramifiedness; the comparisons are retained.

Source reference. Original lemma lemma-unramified-formally-unramified: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2829–2837. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-diagonal-unramified-morphism: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2903–2903. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-locally-quasi-finite-separated-representable: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2903–2903. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-monomorphism-loc-finite-type-loc-quasi-finite: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2903–2903. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma spaces-morphisms-lemma-universally-injective: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 2903–2903. Cited by the retained statement or proof but not part of the collected spans;

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Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. The following are equivalent:*

(1) *f is unramified and a monomorphism,*
(2) *f is unramified and universally injective,*
(3) *f is locally of finite type and a monomorphism,*
(4) *f is universally injective, locally of finite type, and formally unramified.*

Moreover, in this case f is also representable, separated, and locally quasi-finite.

Proof. We have seen in Lemma lemma-unramified-formally-unramified that being formally unramified and locally of finite type is the same thing as being unramified. Hence (4) is equivalent to (2). A monomorphism is certainly formally unramified hence (3) implies (4). It is clear that (1) implies (3). Finally, if (2) holds, then $\Delta : X \to X \times_Y X$ is both an open immersion (Morphisms of Spaces, Lemma spaces-morphisms-lemma-diagonal-unramified-morphism) and surjective (Morphisms of Spaces, Lemma spaces-morphisms-lemma-universally-injective) hence an isomorphism, i.e., f is a monomorphism. In this way we see that (2) implies (1). Finally, we see that f is representable, separated, and locally quasi-finite by Morphisms of Spaces, Lemmas spaces-morphisms-lemma-monomorphism-loc-finite-type-loc-quasi-finite and spaces-morphisms-lemma-locally-quasi-finite-separated-representable. □